

Measure Space: $(\Omega, \mathcal{A}, \mu)$

Ω = Sample space

$\mathcal{A}$ = σ -algebra

μ = Measure

Closure properties:
 $\emptyset, \Omega \in \mathcal{A}$
Complements
Countable unions

Countably additive:
 $\mu(\bigcup A_i) = \sum \mu(A_i)$

Measurable sets $A \in \mathcal{A}$

Measurable functions $f: \Omega \rightarrow \mathbb{R}$
 $f^{-1}(U) \in \mathcal{A}$ for open U

